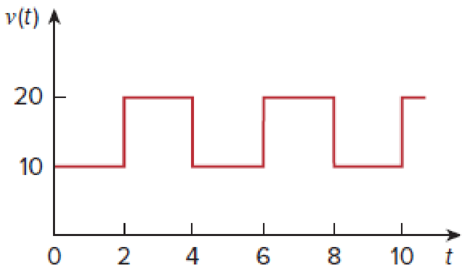
Chapter 11, Problem 26P

Problem

Find the effective value of the voltage waveform in Fig. 11.57.

Figure 11.57:



Step-by-step solution

Step 1 of 2

Consider the following voltage wave form:

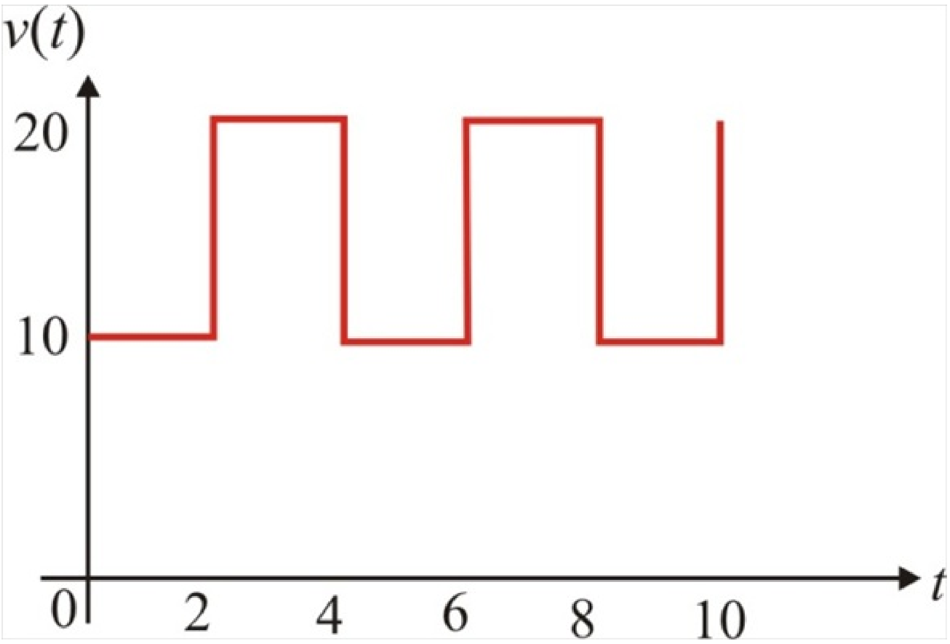


Figure 1

Step 2 of 2

The time period of waveform from Figure 1 is $T = 4$.

Write the expression for the voltage waveform:

$$v(t) = \begin{cases} 10, & 0 < t < 2 \\ 20, & 2 < t < 4 \end{cases}$$

Calculate the rms value of voltage $\mathbf{V}_{\text{rms}}$:

$$\begin{aligned} \mathbf{V}_{\text{rms}} &= \sqrt{\frac{1}{T}\int_0^T v^2 dt} \\ &= \sqrt{\frac{1}{4}\left(\int_0^2 (10)^2\, dt + \int_2^4 (20)^2\, dt\right)} \\ &= \sqrt{\frac{1}{4}\left(100(2-0) + 400(4-2)\right)} \\ &= 15.81 \text{ V} \end{aligned}$$

Therefore, the effective value of voltage $\mathbf{V}_{\text{rms}}$ is $\boxed{15.81 \text{ V}}$.